THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** $6x^2 + 19x + 10$	Multiply term by term and collect like terms
2	**(d)** 63	One part $= 9$
3	**(c)** 1200	CP = SP×100/(100+5)
4	**(a)** 24	Height splits the base: $h^2 = 30^2 - 18^2 = 576$
5	**(a)** 2:3	A:B:C $= 6 : 8 : 9$
6	**(a)** $4x^2 - 12xy + 9y^2$	$(a-b)^2 = a^2 - 2ab + b^2$
7	**(c)** 24	Subtract the added number
8	**(c)** 1800	CP = SP×100/(100−15)
9	**(b)** 84	Other side $= 24$; perimeter $= 2(18 + 24)$
10	**(a)** 24	Total work $= 12 \times 8 \times 15 = 1440$ man-hours
11	**(c)** 2600	$(63 + 37)(63 - 37) = 100 \times 26$
12	**(c)** 18 L	$\frac{3}{10} \times 60$
13	**(b)** 1800	MP = SP×100/(100−15)
14	**(b)** 48	Half of other diagonal $= \sqrt{30^2 - 18^2} = 24$
15	**(a)** 720	One part $= 1350 \div 15 = 90$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$(2x)^2 + 2(2x)(5y) + (5y)^2$ (1) $= 4x^2 + 20xy + 25y^2$ (1).	1+1
17	One part $= 98 \div 14 = 7$ (1); numbers 35 and 63 (1).	1+1
18	CP = SP×100/(100+15) (1) = ₹ 1200 (1).	1+1
19	Height bisects the base, $h^2 = 30^2 - 18^2$ (1); $h = 24$ cm (1).	1+1
20	One part $= 143 \div 13 = 11$ (1); numbers 55 and 88 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$6x^2 - 15x + 4x - 10$ (2); $= 6x^2 - 11x - 10$ (1).	2+1
22	Milk 35 L, water 5 L (1); new water $= \frac{3}{7} \times 35 = 15$ L (1); add 10 L (1).	1+1+1
23	After first discount: ₹ 3600 (1); SP = ₹ 3420 (1); equivalent discount = 14.5% (1).	1+1+1
24	Initial foot distance $= \sqrt{25^2 - 24^2} = 7$ m (1); new height $= \sqrt{25^2 - 15^2} = 20$ m (1); slip $= 24 - 20 = 4$ m (1).	1+1+1
	OR: Total work $= 10 \times 6 \times 24 = 1440$ man-hours (1); workers' hours per day $= 12 \times 8 = 96$ (1); days $= 15$ (1).	1+1+1
25	$197 = 200 - 3$ (1); $(200 - 3)^2 = 40000 - 1200 + 9$ (1); $= 38809$ (1).	1+1+1
	OR: A : B : C $= 3 : 6 : 5$ (1); one part $= 10$ (1); numbers 30, 60, 50 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Plot $= (3x + 2)^2 = 9x^2 + 12x + 4$ (2); pond $= 4x^2$ (1); remaining $= 5x^2 + 12x + 4$ (1); at $x = 10$: 624 m^2 (1).	2+1+1+1
	OR: Total work $= 12 \times 15 = 180$ man-days (1); (i) $180 \div 18 = 10$ days (2); (ii) $180 \div 9 = 20$ men (2).	1+2+2
27	CP = 2100×100/105 = ₹ 2000 (2); new SP − CP = ₹ 400 (1); percent = 20% gain (2).	2+1+2
	OR: (i) $\sqrt{18^2 + 24^2} = 30$ m (2); (ii) saved $= 18 + 24 - 30 = 12$ m (1); (iii) total $= 18 + 24 + 30 = 72$ m (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 100	$(7+3)^2$
2	**(c)** 15 hours	More pipes, less time
3	**(d)** 3000	P = SI×100/(R×T)
4	**(a)** 30	Height difference 18 m; $\sqrt{18^2+24^2}=30$
5	**(c)** 30	Quantity × price is constant
6	**(b)** 54	$(4+2)(4+5)$
7	**(d)** 55	A : B : C = 4 : 6 : 5, so one part is 11
8	**(d)** 240	Original = result × 100/125
9	**(a)** 216	Other leg $=24$; area $=\frac{1}{2}ab$
10	**(a)** 12	Distance $=540$ km
11	**(a)** $8x$	The x^2 and constant terms cancel
12	**(c)** 2 km	1 km = 100000 cm
13	**(a)** 8	Difference = P(R/100)² = 5000×(4/100)²
14	**(a)** 33, 44, 55	Multiples of $(3,4,5)$ by 11 stay a triplet
15	**(d)** 32	One part $=8$; difference $=(11-7)\times 8$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$39\times 41=(40)^2-(1)^2$ (1) $=1600-1=1599$ (1).	1+1
17	45 km/h $=45\times\frac{1000}{3600}$ (1) $=12.5$ m/s (1).	1+1
18	SI = ₹ 840 (1); R = SI×100/(P×T) = 5% (1).	1+1
19	Other side $=32$ cm (1); area $=768$ cm² (1).	1+1
20	Student-days $=50\times 16=800$ (1); days $=800\div 40=20$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	SI = ₹ 1800 (1); T = SI×100/(P×R) (1) = 3 years (1).	1+1+1
22	$18^2+24^2=900=30^2$ (1); right-angled with legs 18, 24 (1); area $=\frac{1}{2}\times 18\times 24=216$ cm² (1).	1+1+1
23	A:B:C $=3:6:8$ (1); one part $=10$ (1); A = 30, B = 60, C = 80 (1).	1+1+1
24	LHS $=49+9=58$ (1.5); RHS $=2(25+4)=58$ (1.5).	1½+1½
	OR: One part $=6$; daughter 18 (1); 4 years ago: 38 and 14 (1); ratio $19:7$ (1).	1+1+1
25	After year 1: 10000×110/100 = ₹ 11000 (1); after year 2: ×120/100 = ₹ 13200 (2).	1+2
	OR: Hypotenuse $=30$ cm (1); area $=\frac{1}{2}\times 18\times 24=216$ (1); $\frac{1}{2}\times 30\times h=216$ gives $h=14.4$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Milk $=24$ L, water $=36$ L (2); new water $=48$ L (1); water to add $=12$ L (2).	2+1+2
	OR: (i) $103^2=10609$ (1.5); (ii) $88\times 92=8096$ (1.5); (iii) $(76+24)(76-24)=100\times 52=5200$ (2).	1½+1½+2
27	Ratio $3:5$ (1); one part ₹ 100 (1); Anil ₹ 300 (1.5); Bhanu ₹ 500 (1.5).	1+1+1½+1½
	OR: Total sum = ₹ 10000 (1); R = SI×100/(sum×T) = 5% (2); interest on first = ₹ 400 (1); on second = ₹ 600 (1).	1+2+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | Class 8 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** $2x + 22$	Expand each bracket and collect like terms
2	**(d)** 10 m/s	Multiply by $\frac{5}{18}$
3	**(d)** 1620	2000×90/100×90/100
4	**(c)** 36	Height splits the base: $h^2 = 39^2 - 15^2 = 1296$
5	**(d)** 32	$80/360 \times 144$
6	**(c)** $12x^2 + 11x + 2$	Multiply term by term and collect like terms
7	**(c)** 48	One part $= 6$
8	**(a)** 2000	CP = SP×100/(100+20)
9	**(a)** 102	Other side $= 36$; perimeter $= 2(15 + 36)$
10	**(d)** 4:5	A:B:C $= 4 : 14 : 5$
11	**(b)** $9x^2 - 6xy + y^2$	$(a - b)^2 = a^2 - 2ab + b^2$
12	**(d)** 12	Subtract the added number
13	**(a)** 3500	CP = SP×100/(100−20)
14	**(d)** 72	Half of other diagonal $= \sqrt{39^2 - 15^2} = 36$
15	**(d)** 18	Total work $= 9 \times 6 \times 20 = 1080$ man-hours

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$9x + 27 - 4x + 8$ (1) $= 5x + 35$ (1).	1+1
17	Actual distance $= 6 \times 50000$ cm $= 300000$ cm (1); $= 3$ km (1).	1+1
18	Increase = ₹ 40 (1); % = 40/160×100 = 25% (1).	1+1
19	Half of other diagonal$^2 = 50^2 - 30^2 = 1600$ (1); other diagonal $= 80$ cm (1).	1+1
20	Students $= 120/360 \times 108 = 36$ (1); percent $= 33.33\%$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Students per degree $= 15/60$ (1); total $= 15 \times 360/60 = 90$ (1); 96° sector $= 24$ students (1).	1+1+1
22	$12x^2 - 8x + 3x - 2$ (2); $= 12x^2 - 5x - 2$ (1).	2+1
23	Milk 36 L, water 24 L (1); new water $= \frac{4}{3} \times 36 = 48$ L (1); add 24 L (1).	1+1+1
24	After first discount: ₹ 2000 (1); SP = ₹ 1800 (1); equivalent discount = 28% (1).	1+1+1
	OR: Initial foot distance $= \sqrt{25^2 - 20^2} = 15$ m (1); new height $= \sqrt{25^2 - 20^2} = 15$ m (1); slip $= 20 - 15 = 5$ m (1).	1+1+1
25	Total work $= 15 \times 8 \times 12 = 1440$ man-hours (1); workers' hours per day $= 18 \times 5 = 90$ (1); days $= 16$ (1).	1+1+1
	OR: $108 = 110 - 2$ (1); $(110 - 2)^2 = 12100 - 440 + 4$ (1); $= 11664$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Height$^2 = 40^2 - 24^2 = 1024$, height $= 32$ cm (2); area $= \frac{1}{2} \times 48 \times 32 = 768$ cm² (2); perimeter $= 2 \times 40 + 48 = 128$ cm (1).	2+2+1
	OR: Work done in 8 days $= 8/20$ of the task; remaining $= \frac{3}{5}$ (1); workers now $= 18$; remaining man-days $= 144$ (2); days $= (12 \times 12) \div 18 = 8$ (2).	1+2+2
27	Plot $= (4x + 5)^2 = 16x^2 + 40x + 25$ (2); pond $= 4x^2$ (1); remaining $= 12x^2 + 40x + 25$ (1); at $x = 10$: 1625 m² (1).	2+1+1+1
	OR: Total work $= 20 \times 24 = 480$ man-days (1); (i) $480 \div 16 = 30$ days (2); (ii) $480 \div 30 = 16$ men (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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